OVER A FLAT NEIGHBOURHOOD COVERING THE I-POWER TORSION MODULES ON BOTH SIDES ARE EQUIVALENT AND HOM DOES NOT CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes the union symbol in the exponent as I^\infty throughout. The counted proof span ends at source line 24360, which is the author's \end{proof}; the paragraph proving essential surjectivity in part (3) is inside it. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- I-power torsion submodules and the quotient by the torsion submodule. used at proof lines 24326-24331. Is the reduction that removes the torsion part and leaves a torsion-free quotient to control.
- The adjoint tensor-restrict isomorphism for a base change of rings. invoked at proof lines 24346-24347. Gives the Hom isomorphism in the case where the target module is already I-power torsion.
- The neighbourhood isomorphism lemma identifying modules with their base change when annihilated by a power of I. invoked at proof line 24330 and 24351-24352. Lets M[I^∞] and its base change be identified, and later lets N be replaced by N[I^∞].
- The divide-by-torsion lemma producing a module with no I-power torsion. invoked at proof line 24333. Supplies M'[I^∞] = 0, which makes the map into the direct sum of t copies injective.

Source reference. Original lemma 05DQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6963–6963. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 24227: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24227–24227. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 24042: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24042–24042. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 24264: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24264–24264. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Assume $\varphi : R \to S$ is a flat ring map and $I \subset R$ is a finitely generated ideal such that $R/I \to S/IS$ is an isomorphism. Then*

1. *for any R-module M the map $M \to M \otimes_R S$ induces an isomorphism $M[I^\infty] \to (M \otimes_R S)[(IS)^\infty]$ of I-power torsion submodules,*
2. *the natural map*
$$\mathrm{Hom}_R(M, N) \longrightarrow \mathrm{Hom}_S(M \otimes_R S, N \otimes_R S)$$
is an isomorphism if either M or N is I-power torsion, and
3. *the base change functor $M \mapsto M \otimes_R S$ defines an equivalence of categories between I-power torsion modules and IS-power torsion modules.*

Proof. Note that the equivalent conditions of both Lemma 24227 and Lemma 24264 are satisfied. We will use these without further mention. We first prove (1). Let M be any R-module. Set $M' = M/M[I^\infty]$ and consider the exact sequence
$$0 \to M[I^\infty] \to M \to M' \to 0$$
As $M[I^\infty] = M[I^\infty] \otimes_R S$ we see that it suffices to show that $(M' \otimes_R S)[(IS)^\infty] = 0$. Write $I = (f_1, \ldots, f_t)$. By Lemma 24042 we see that $M'[I^\infty] = 0$. Hence for every $n > 0$ the map
$$M' \longrightarrow \bigoplus\nolimits_{i=1,\ldots t} M', \quad x \longmapsto (f_1^n x, \ldots, f_t^n x)$$
is injective. As S is flat over R also the corresponding map $M' \otimes_R S \to \bigoplus_{i=1,\ldots t} M' \otimes_R S$ is injective. This means that $(M' \otimes_R S)[I^n] = 0$ as desired.

Next we prove (2). If N is I-power torsion, then $N \otimes_R S = N$ and the displayed map of (2) is an isomorphism by Algebra, Lemma 05DQ. If M is I-power torsion, then the image of any map $M \to N$ factors through $N[I^\infty]$ and the image of any map $M \otimes_R S \to N \otimes_R S$ factors through $(N \otimes_R S)[(IS)^\infty]$. Hence in this case part (1) guarantees that we may replace N by $N[I^\infty]$ and the result follows from the case where N is I-power torsion we just discussed.

Next we prove (3). The functor is fully faithful by (2). For essential surjectivity, we simply note that for any IS-power torsion S-module N, the natural map $N \otimes_R S \to N$ is an isomorphism. □